

ATLANTIC, IA, USA

WMO: 725453

Lat: 41.407N

Lon: 95.047W

Elev: 1182

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 14930

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	0.0	-14.9	2.6	-5.5	-8.9	3.6	0.9	31.2	16.8	27.5	22.7	5.4	0	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	93.0	77.2	90.5	76.2	88.1	75.0	80.5	89.4	78.4	87.1	76.6	85.1	11.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	154.7	87.9	75.4	138.9	84.1	73.3	129.2	82.1	45.2	90.4	43.0	87.8	40.8	85.7	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.0	23.1	19.8	-15.9	96.6	4.8	3.3	-19.3	99.0	-22.2	100.9	-24.9	102.8	-28.4	105.2		
			-16.2	82.9	4.2	2.0	-19.3	84.3	-21.8	85.5	-24.2	86.6	-27.3	88.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	23.0	26.5	38.7	50.3	61.7	72.1	75.3	72.0	65.2	52.1	38.1	26.6
	DBStd	20.90	12.41	12.77	12.33	10.28	8.78	6.29	6.10	5.73	8.68	9.83	10.12	11.44
	HDD50	3209	837	659	390	118	10	0	0	3	93	373	726	
	HDD65	6360	1302	1078	818	451	168	13	4	10	105	415	806	1190
	CDD50	3304	0	1	38	128	374	662	785	682	459	157	17	1
	CDD65	982	0	0	2	11	67	225	324	227	111	15	0	0
	CDH74	10097	0	0	22	155	715	2308	3460	2122	1152	159	4	0
	CDH80	3766	0	0	2	38	232	862	1446	754	396	36	0	0
Wind	WSAvg	8.5	9.2	9.3	9.9	10.9	9.7	8.1	6.2	5.6	7.0	8.4	9.0	8.6
Precipitation	PrecAvg	30.0	0.7	0.9	2.2	3.1	3.8	4.0	3.7	3.7	4.1	2.5	1.4	1.1
	PrecMax	43.7	2.2	2.4	6.0	7.7	7.2	11.1	8.5	10.6	13.2	4.8	4.1	3.5
	PrecMin	19.2	0.0	0.1	0.2	0.9	1.6	0.7	0.4	0.3	0.6	0.1	0.0	0.2
	PrecStd	6.0	0.6	0.7	1.6	1.8	1.7	2.8	1.9	2.3	3.0	1.5	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	63.3	77.0	86.1	92.5	94.5	97.2	95.4	92.9	84.3	72.3	60.6
		MCWB	44.6	52.2	60.1	64.6	72.0	74.4	80.7	79.1	73.8	67.2	58.7	54.3
	2%	DB	48.1	55.2	70.3	78.9	86.1	90.8	93.2	90.7	88.5	78.9	64.2	52.3
		MCWB	42.2	47.4	57.7	61.3	70.1	75.0	79.6	76.8	73.3	64.9	55.1	47.5
	5%	DB	43.2	49.5	63.7	73.1	81.4	88.2	90.7	87.6	84.3	73.1	59.1	46.4
		MCWB	38.0	44.2	54.3	58.8	66.0	73.7	78.5	76.0	72.0	61.2	50.9	41.3
	10%	DB	38.6	43.2	58.6	68.4	77.2	85.6	88.1	83.9	81.1	69.6	54.5	41.5
		MCWB	35.1	38.6	49.9	56.6	64.8	72.3	77.2	73.9	69.1	59.0	47.4	37.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	54.8	64.1	67.5	75.0	79.3	84.3	82.1	78.1	72.6	61.0	56.0
		MCDB	52.5	60.8	73.4	78.5	86.2	88.7	93.8	90.9	88.0	80.1	68.6	59.4
	2%	WB	42.3	48.2	59.2	64.1	71.3	77.2	82.1	79.2	75.5	67.1	56.3	47.6
		MCDB	48.1	55.5	66.2	75.4	82.9	86.9	91.2	88.1	84.3	74.4	63.9	50.7
	5%	WB	38.6	43.3	55.1	60.9	68.7	75.4	79.8	77.0	73.6	63.6	51.9	41.8
		MCDB	43.0	49.1	63.4	70.8	78.5	85.0	88.4	85.6	81.8	69.7	58.2	46.3
	10%	WB	35.4	39.1	50.0	58.1	66.5	73.8	77.9	74.7	71.5	60.1	47.7	38.2
		MCDB	38.7	43.3	58.5	67.0	75.2	83.2	85.9	82.2	79.4	67.6	54.2	42.1
Mean Daily Temperature Range	5% DB	MDBR	17.8	18.3	20.4	22.7	21.0	20.4	20.4	20.7	24.2	22.7	20.5	17.3
		MCDBR	23.9	25.2	29.5	31.0	26.5	22.9	22.2	24.4	26.4	28.1	28.6	24.2
		MCWBR	18.5	18.2	18.7	17.8	13.5	11.7	12.6	14.1	14.2	16.5	19.3	18.5
	5% WB	MCDBR	22.7	23.6	25.6	24.7	21.8	19.8	19.8	21.2	22.6	22.8	26.0	22.3
		MCWBR	17.9	17.8	17.3	16.0	12.3	12.0	13.1	13.7	13.3	15.9	19.1	17.9
Clear-Sky Solar Irradiance	taub	0.284	0.311	0.326	0.382	0.406	0.413	0.430	0.417	0.380	0.334	0.295	0.284	
	taud	2.424	2.365	2.395	2.258	2.264	2.298	2.239	2.287	2.345	2.462	2.499	2.443	
	Ebn at Noon	278	285	293	281	276	274	267	268	271	273	272	268	
	Edh at Noon	25	31	34	42	43	41	43	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	624	935	1206	1525	1817	2014	2033	1758	1463	1006	692	524	
	RadStd	56	57	112	112	140	172	144	117	100	120	52	50	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page